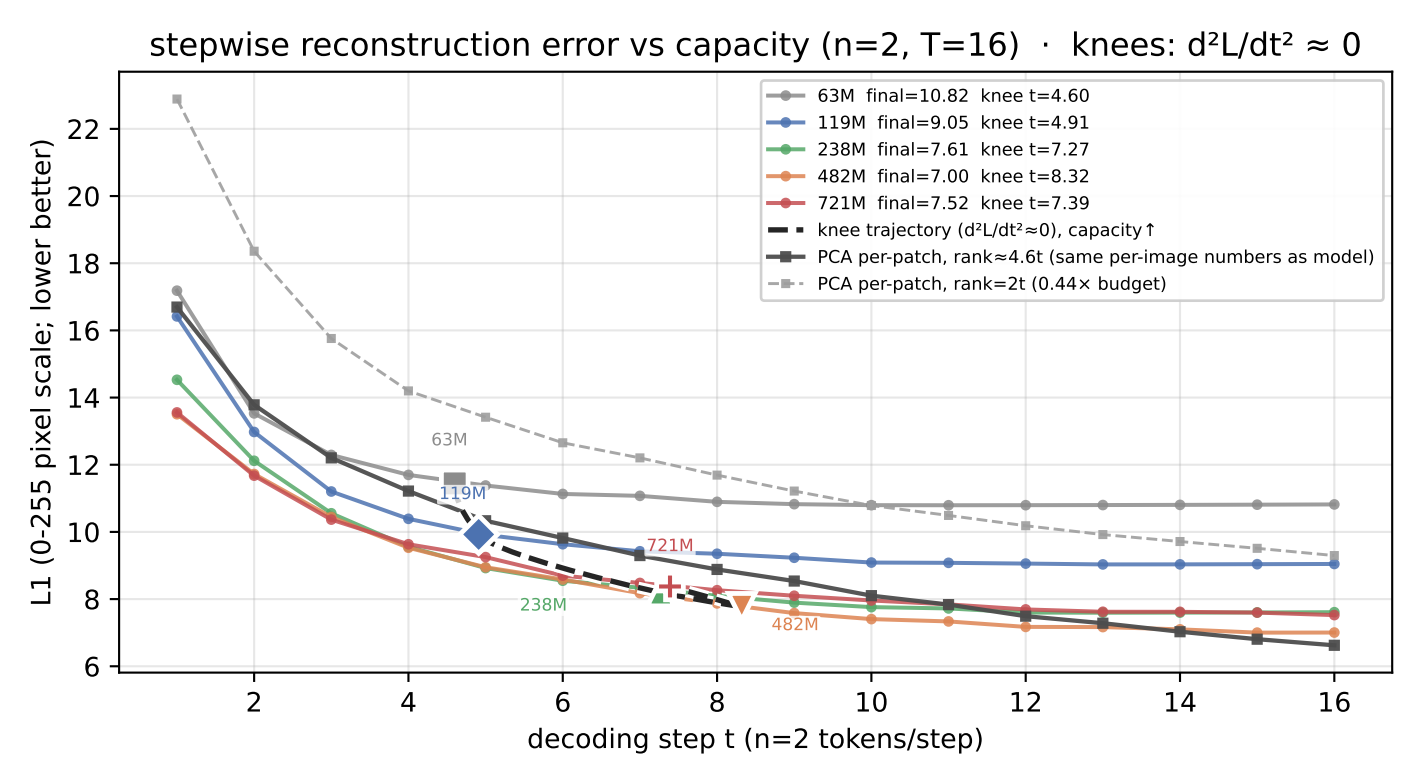

knee = d²L/dt² ≈ 0: first t∈[1.5,15.5] with |d²L/dt²| ≤ 10% of its peak (≈ second derivative decayed to zero).
smooth: Savitzky–Golay (w=5, p=2) → UnivariateSpline (s=0.001·n) → d²L/dt² on a 0.01-step grid; elbow (max|d²L/dt²|) is only a cross-check (t≈1.5–2.0).
All five curves are monotone convex decays → no strict interior d²L/dt²=0; the 63M baseline has one strict crossing at t≈13.1 (tail upturn), not marked.
budget: model step t = 896t numbers (2t tokens × 448); PCA rank r = 196r numbers/image (+ 768×160 shared basis learned on
train patches, amortized ≈25/img over 5,000). PCA = per-patch PCA on the same 5,000-image val set, not per-image oracle.